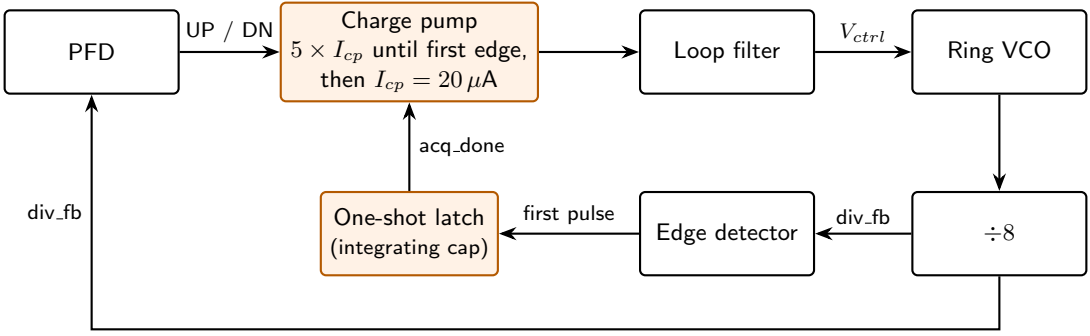

PFD
UP / DN
Charge pump
$5 \times I_{cp}$ until first edge,
then $I_{cp} = 20\,\mu$A
Loop filter
V_{ctrl}
Ring VCO
acq_done
div_fb
One-shot latch
(integrating cap)
first pulse
Edge detector
div_fb
$\div 8$